

SPAM EMAIL CLASSIFICATION USING DECISION TREES AND XGBOOST WITH CROSS-VALIDATION

ABSTRACT

Spam emails are unwanted messages that may contain advertisements, phishing attempts, fraudulent content, or malicious links. The large volume of emails received by users makes manual identification of spam difficult. Machine learning can be used to automatically classify emails based on patterns learned from previously labeled messages.

This project presents a machine learning-based approach for **spam email classification using the CEAS-08 dataset**. The proposed workflow consists of data cleaning, Exploratory Data Analysis (EDA), TF-IDF feature extraction, train-test splitting, model training, 5-Fold Stratified Cross-Validation, and performance evaluation.

Two tree-based machine learning algorithms, **Decision Tree and XGBoost**, were implemented and compared. TF-IDF was used to convert email text into numerical features using unigrams and bigrams. The models were evaluated using Accuracy, Precision, Recall, and F1 Score. Cross-validation was also performed to examine the consistency of the models.

The experimental results show that both models achieved high classification performance. However, **XGBoost achieved better overall results**, with an Accuracy of **99.14%**, Precision of **98.82%**, Recall of **99.66%**, and F1 Score of **99.24%**. Its 5-Fold Cross-Validation F1 Score was **99.15%**, compared with **98.54%** for Decision Tree.

1. INTRODUCTION

Email is one of the most commonly used forms of digital communication. Along with useful communication, users receive a large number of unwanted emails known as spam. Spam emails can contain advertisements, phishing attempts, fraudulent messages, and potentially harmful links.

As the number of emails increases, manually identifying spam becomes inefficient. Therefore, automated spam classification systems are useful for identifying unwanted messages before they reach the user.

Machine learning provides an effective approach to this problem because models can learn patterns from previously classified emails and use those patterns to classify new messages.

In this project, the **CEAS-08 dataset** is used to develop a spam email classification system. The project compares two machine learning algorithms, **Decision Tree and XGBoost**, using textual email features extracted through TF-IDF. Their performance is evaluated using standard classification metrics and 5-Fold Stratified Cross-Validation.

2. PROBLEM STATEMENT

The objective of this project is to develop an automated machine learning system that can classify emails as **spam or legitimate** based on their textual content.

The system should:

- Process and clean raw email data.
 - Analyze the characteristics of the dataset.
 - Extract meaningful features from email text.
 - Train machine learning classification models.
 - Evaluate model performance using multiple metrics.
 - Validate model consistency using cross-validation.
 - Compare Decision Tree and XGBoost performance.
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3. OBJECTIVES

The main objectives of the project are:

1. To preprocess and clean the CEAS-08 email dataset.
 2. To perform Exploratory Data Analysis on the email data.
 3. To extract useful textual features using TF-IDF.
 4. To implement a Decision Tree classification model.
 5. To implement an XGBoost classification model.
 6. To evaluate both models using Accuracy, Precision, Recall, and F1 Score.
 7. To perform 5-Fold Stratified Cross-Validation.
 8. To compare the performance and consistency of both models.
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4. DATASET DESCRIPTION

The project uses the **CEAS-08 dataset** for spam email classification.

The dataset contains email-related information, including textual fields such as the **subject and body**. The target variable used for classification is the email label.

Classes

- **0 – Legitimate Email**
- **1 – Spam Email**

Dataset Summary

Category	Number of Emails
Total Emails	39,154

Category	Number of Emails
Legitimate Emails	17,312
Spam Emails	21,842

The subject and body fields were primarily used as textual input for classification.

5. DATA PREPROCESSING

Raw email data can contain missing values, duplicate records, HTML tags, URLs, email addresses, numbers, and other unnecessary characters. Therefore, preprocessing was performed before applying machine learning algorithms.

5.1 Handling Missing Values

Missing values in important text fields such as the subject and body were identified and handled so that they would not interfere with further processing.

5.2 Removing Duplicate Records

Duplicate records were checked and removed to prevent repeated emails from unnecessarily influencing the model.

5.3 Text Cleaning

The email text was cleaned by removing unnecessary elements such as:

- HTML tags
- URLs
- Email addresses
- Numbers
- Special characters
- Extra spaces

The text was also converted into a consistent format.

5.4 Combining Subject and Body

The cleaned subject and body were combined into a single text field. This provided the model with more information about the content of each email.

Preprocessing Flow

Raw Email → Missing Value Handling → Duplicate Removal → Text Cleaning → Subject + Body → Cleaned Text

6. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) was performed to understand the dataset before training the machine learning models.

The following characteristics were analyzed:

- Spam and legitimate email distribution
- Email length
- Word count
- Subject length
- Number of URLs

6.1 Class Distribution

The dataset contains:

- **17,312 legitimate emails**
- **21,842 spam emails**

This shows the distribution of the two classes used for classification.

6.2 Email Length

Email length was analyzed to understand whether spam and legitimate emails differ in their amount of textual content.

6.3 Word Count

The number of words in emails was examined to identify differences in message content between the two classes.

6.4 Subject Length

The length of email subjects was analyzed because subject lines can contain useful information for spam classification.

6.5 URL Count

The number of URLs present in emails was also examined because links can be an important characteristic of spam or phishing-related messages.

EDA helped provide an understanding of the dataset before feature extraction and model training.

7. FEATURE EXTRACTION USING TF-IDF

Machine learning models cannot directly process raw text. Therefore, the email text was converted into numerical features using **TF-IDF (Term Frequency-Inverse Document Frequency)**.

TF-IDF assigns importance to words based on their frequency in an email and their occurrence across the collection of emails.

The implementation used:

- **Unigrams** – individual words
- **Bigrams** – combinations of two words
- **English stop-word removal**
- **Maximum 10,000 features**

For example:

Unigrams: free, offer, prize

Bigrams: free offer, click here

This allows the model to capture both individual spam-related words and useful phrases.

The TF-IDF vectorizer was fitted only on the training data and then applied to the test data to avoid data leakage.

Feature Extraction Flow

Email Text → TF-IDF Vectorization → Numerical Feature Matrix → Machine Learning Models

8. TRAIN-TEST SPLIT

The dataset was divided into training and testing data.

- **80%** of the data was used for training.
- **20%** was used for testing.
- A fixed random state was used to make the experiment reproducible.
- **Stratification** was applied to maintain the class distribution.

The training data was used to build the models, while the test data was used to evaluate their performance on unseen emails.

9. MACHINE LEARNING MODELS

Two tree-based machine learning algorithms were used.

9.1 Decision Tree

A Decision Tree is a supervised machine learning algorithm that makes predictions using a sequence of decision rules.

The model divides the feature space into different regions based on learned conditions and finally assigns a class to each input.

In this project, the Decision Tree uses TF-IDF features to classify emails into spam and legitimate categories.

9.2 XGBoost

XGBoost is a gradient boosting algorithm based on an ensemble of decision trees.

It builds multiple trees sequentially, with each new tree helping improve the errors made by previous trees.

In this project, XGBoost was trained using TF-IDF features to classify emails into the two target classes.

10. CROSS-VALIDATION

To examine the consistency of model performance, **5-Fold Stratified Cross-Validation** was performed.

The training data was divided into five subsets. In each iteration:

- Four subsets were used for training.
- One subset was used for validation.
- The process was repeated five times.

The results from the five folds were then averaged.

Stratification ensures that the proportion of spam and legitimate emails remains reasonably consistent across the folds.

Cross-validation provides additional evidence about how consistently a model performs across different portions of the training data.

11. MODEL EVALUATION

The models were evaluated using four main metrics.

11.1 Accuracy

Accuracy represents the percentage of all emails that were classified correctly.

11.2 Precision

Precision indicates how many emails predicted as spam were actually spam.

11.3 Recall

Recall indicates how many of the actual spam emails were correctly detected.

11.4 F1 Score

F1 Score combines Precision and Recall into a single metric and provides a balance between the two.

12. RESULTS AND COMPARISON

The following results were obtained from the test dataset.

12.1 Test-Set Performance

Model	Accuracy	Precision	Recall	F1 Score
Decision Tree	98.70%	98.28%	99.40%	98.84%
XGBoost	99.14%	98.82%	99.66%	99.24%

The results show that both models achieved high performance.

Decision Tree achieved:

- Accuracy: **98.70%**
- Precision: **98.28%**
- Recall: **99.40%**
- F1 Score: **98.84%**

XGBoost achieved:

- Accuracy: **99.14%**
- Precision: **98.82%**
- Recall: **99.66%**
- F1 Score: **99.24%**

XGBoost achieved higher values for all four reported metrics.

13. CROSS-VALIDATION RESULTS

The 5-Fold Cross-Validation results were:

Model	CV F1 Score
Decision Tree	98.54%
XGBoost	99.15%

The higher cross-validation F1 Score of XGBoost indicates that it maintained strong performance across the validation folds.

The difference between the test F1 and CV F1 scores was relatively small for both models, indicating that their performance remained broadly consistent across the evaluation procedure.

14. CONFUSION MATRIX ANALYSIS

Confusion matrices were generated to provide a detailed view of the classification results.

A confusion matrix contains four categories:

- **True Positive (TP):** Spam correctly classified as spam.
- **True Negative (TN):** Legitimate email correctly classified as legitimate.
- **False Positive (FP):** Legitimate email incorrectly classified as spam.
- **False Negative (FN):** Spam incorrectly classified as legitimate.

The confusion matrices help identify the types of errors made by each model and provide more detailed information than accuracy alone.

15. MODEL COMPARISON

The two models can be compared based on their test-set and cross-validation results.

Metric	Decision Tree	XGBoost
Accuracy	98.70%	99.14%
Precision	98.28%	98.82%
Recall	99.40%	99.66%
F1 Score	98.84%	99.24%
CV F1 Score	98.54%	99.15%

Based on the experimental results, **XGBoost achieved higher values across all reported evaluation metrics.**

Its F1 Score was **99.24%**, compared with **98.84%** for Decision Tree. XGBoost also achieved a higher CV F1 Score of **99.15%**, compared with **98.54%** for Decision Tree.

Therefore, for this particular dataset and experimental setup, XGBoost produced the stronger classification results.

16. KEY FINDINGS

The major findings of the project are:

1. The CEAS-08 dataset contained **39,154 emails**, including 17,312 legitimate emails and 21,842 spam emails.
 2. Text preprocessing helped prepare the raw email content for machine learning.
 3. TF-IDF successfully transformed email text into numerical features using unigrams and bigrams.
 4. Both Decision Tree and XGBoost achieved high classification performance.
 5. XGBoost achieved the highest test-set F1 Score of **99.24%** and the highest CV F1 Score of **99.15%**.
 6. XGBoost also achieved higher Accuracy, Precision, and Recall than Decision Tree in the reported test results.
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17. CONCLUSION

This project successfully developed a machine learning pipeline for **spam email classification using the CEAS-08 dataset**. The complete workflow included data preprocessing, Exploratory Data Analysis, TF-IDF feature extraction, train-test splitting, model training, cross-validation, and evaluation.

Both Decision Tree and XGBoost were able to classify spam and legitimate emails with high accuracy. Decision Tree achieved an Accuracy of **98.70%** and an F1 Score of **98.84%**, while XGBoost achieved an Accuracy of **99.14%** and an F1 Score of **99.24%**.

The cross-validation results also showed strong performance, with XGBoost achieving a **99.15% CV F1 Score**, compared with **98.54%** for Decision Tree.

Based on the experimental results, **XGBoost achieved better overall performance than Decision Tree for this spam email classification task**. The results demonstrate that machine learning combined with TF-IDF text representation can be effectively applied to automated spam email classification.

18. FUTURE SCOPE

The project can be extended in several ways:

1. **Hyperparameter Optimization:** Different model parameters can be tested to further improve performance.
2. **Advanced NLP Techniques:** Word embeddings and transformer-based models can be explored for richer text representations.
3. **Additional Features:** Sender information, URL characteristics, email headers, and other metadata can be incorporated.

4. **Real-Time Classification:** The trained model can be integrated into a real-time email filtering system.
 5. **Web Application:** A user-friendly web interface can be developed where users can enter or upload an email and receive a spam/legitimate prediction.
 6. **Model Deployment:** The final model can be deployed as an API or integrated with an email service.
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19. REFERENCES

1. CEAS-08 Email Dataset, Kaggle.
 2. Scikit-learn documentation for TF-IDF and machine learning algorithms.
 3. XGBoost documentation for gradient boosting and classification.
 4. Research literature related to machine learning-based spam email classification.
 5. Research literature related to text classification and TF-IDF feature extraction.
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20. PROJECT WORKFLOW

The complete workflow of the proposed system is:

CEAS-08 Dataset

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Data Cleaning

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Exploratory Data Analysis

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TF-IDF Feature Extraction

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Train-Test Split

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Decision Tree + XGBoost

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5-Fold Stratified Cross-Validation

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Accuracy / Precision / Recall / F1 Evaluation

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Model Comparison

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Final Conclusion